

SQL Solutions Reference

TARGET TABLE CONTEXT: EMPLOYEESALES

EmployeeSales (SaleID INT, EmployeeID INT, Department VARCHAR(50), SaleAmount DECIMAL(10,2), SaleDate DATE)

Q1. Write a query to calculate the total sales amount for each department in the EmployeeSales table.

```
SELECT Department, SUM(SaleAmount) AS [Total Sales]
FROM EmployeeSales
GROUP BY Department;
```

Q2. Write a query to count the number of sales made by each employee.

```
SELECT EmployeeID, COUNT(EmployeeID) AS [No Of Sales]
FROM EmployeeSales
GROUP BY EmployeeID;
```

Q3. Write a query to calculate the average sale amount for each department.

```
SELECT Department, AVG(SaleAmount) AS [Average Sales]
FROM EmployeeSales
GROUP BY Department;
```

Q4. Write a query to find the total sales amount for each employee, but only include employees who have made more than one sale.

```
SELECT EmployeeID, SUM(SaleAmount) AS [Total Sales]
FROM EmployeeSales
GROUP BY EmployeeID
HAVING COUNT(EmployeeID) > 1;
```

Q5. Write a query to find the total sales for each month in 2023.

```
SELECT MONTH(SaleDate) AS Month, SUM(SaleAmount) AS [Total Sales]
FROM EmployeeSales
WHERE YEAR(SaleDate) = 2023
GROUP BY MONTH(SaleDate)
ORDER BY Month;
```

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Assignment 02**TARGET TABLE CONTEXT: EMPLOYEES**

Employees (EmployeeID INT, FirstName NVARCHAR, LastName NVARCHAR, Email NVARCHAR, DepartmentID INT, HireDate DATE, Salary DECIMAL)

Q1. Write a SQL query to find the names of employees who have a salary higher than the average salary of all employees.

```
SELECT FirstName, Salary
FROM Employees
WHERE Salary > (
    SELECT AVG(Salary)
    FROM Employees
);
```

Q2. Write a SQL query to list the employee names and their departments for employees who were hired after the oldest employee in the company.

```
SELECT FirstName, LastName, DepartmentID
FROM Employees
WHERE HireDate > (
    SELECT MIN(HireDate)
    FROM Employees
);
```

Q3. Write a SQL query to find the details of the employee(s) with the highest salary.

```
SELECT *
FROM Employees
WHERE Salary = (
    SELECT MAX(Salary)
    FROM Employees
);
```

Q4. Write a SQL query to find the names of employees who work in the same department as 'John Smith'.

```
SELECT FirstName, LastName
FROM Employees
WHERE DepartmentID = (
    SELECT DepartmentID FROM Employees WHERE FirstName = 'John' AND LastName = 'Smith'
)
AND NOT (FirstName = 'John' AND LastName = 'Smith');
```

Q5. Write a SQL query to find the names of employees who do not belong to the department with the highest average salary.

```
SELECT FirstName, LastName
FROM Employees
WHERE DepartmentID <> (
    SELECT TOP 1 DepartmentID FROM Employees
    GROUP BY DepartmentID ORDER BY AVG(Salary) DESC
);
```

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Assignment 03**TARGET TABLE CONTEXT: EMPLOYEES**

Employees (EmployeeID INT, FirstName NVARCHAR, LastName NVARCHAR, Email NVARCHAR, DepartmentID INT, HireDate DATE, Salary DECIMAL)

Q1. Write a SQL query to find the names of employees whose first names start with the letter 'J'.

```
SELECT FirstName, LastName
FROM Employees
WHERE FirstName LIKE 'J%';
```

Q2. Write a SQL query to find the names of employees whose last names end with the letter 'n'.

```
SELECT FirstName, LastName
FROM Employees
WHERE LastName LIKE '%n';
```

Q3. Write a SQL query to find the email addresses of employees that contain the word "john".

```
SELECT Email
FROM Employees
WHERE Email LIKE '%john%';
```

Q4. Write a SQL query to find the names of employees whose first names have exactly 5 characters.

```
SELECT FirstName, LastName
FROM Employees
WHERE FirstName LIKE '_____';
```

Q5. Write a SQL query to find the names of employees whose last names contain the letter 'a' as the second character.

```
SELECT FirstName, LastName
FROM Employees
WHERE LastName LIKE '_a%';
```

TARGET TABLES CONTEXT: EMPLOYEES & DEPARTMENTS

Employees (EmployeeID, FirstName, LastName, Email, DepartmentID, HireDate, Salary)

Departments (DepartmentID INT, DepartmentName NVARCHAR)

Q1. Write a SQL query to list the names of employees along with the names of the departments they work in.

```
SELECT e.FirstName, e.LastName, d.DepartmentName
FROM Employees e
JOIN Departments d ON e.DepartmentID = d.DepartmentID;
```

Q2. Write a SQL query to list all the departments and the employees working in them, including departments with no employees.

```
SELECT d.DepartmentName, e.FirstName, e.LastName
FROM Departments d
LEFT JOIN Employees e ON d.DepartmentID = e.DepartmentID;
```

Q3. Write a SQL query to find the names of employees who do not belong to any department (i.e., no matching department ID).

```
SELECT e.FirstName, e.LastName
FROM Employees e
LEFT JOIN Departments d ON e.DepartmentID = d.DepartmentID
WHERE d.DepartmentID IS NULL;
```

Q4. Write a SQL query to list the names of employees who work in the same department as 'Jane Doe'.

```
SELECT e.FirstName, e.LastName
FROM Employees e
JOIN Employees j ON e.DepartmentID = j.DepartmentID
WHERE j.FirstName = 'Jane' AND j.LastName = 'Doe' AND e.EmployeeID <> j.EmployeeID;
```

Q5. Write a SQL query to find the department with the highest total salary paid to its employees.

```
SELECT TOP 1 d.DepartmentName, SUM(e.Salary) AS Total_Salary
FROM Departments d
JOIN Employees e ON d.DepartmentID = e.DepartmentID
GROUP BY d.DepartmentName
ORDER BY SUM(e.Salary) DESC;
```

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Assignment 05

TARGET TABLE CONTEXT: EMPLOYEES

Employees (EmployeeID INT, FirstName NVARCHAR, LastName NVARCHAR, Email NVARCHAR, DepartmentID INT, HireDate DATE, Salary DECIMAL)

Q1. Write a SQL query to select all columns and rows from the Employees table.

```
SELECT * FROM Employees;
```

Q2. Write a SQL query to find the names and email addresses of all employees who work in the department with DepartmentID = 101.

```
SELECT FirstName, LastName, Email
FROM Employees
WHERE DepartmentID = 101;
```

Q3. Write a SQL query to find the total number of employees in the Employees table.

```
SELECT COUNT(EmployeeID) AS [Total_Employees]
FROM Employees;
```

Q4. Write a SQL query to find the details of employees who were hired in the year 2020.

```
SELECT *
FROM Employees
WHERE YEAR(HireDate) = 2020;
```

Q5. Write a SQL query to update the salary of 'Jane Doe' to 90,000.

```
SELECT * INTO temp FROM Employees;

UPDATE temp
SET Salary = 90000
WHERE FirstName = 'Jane' AND LastName = 'Doe';
```